

# LOS + LMS Prototype - Low-Level Design

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LMS Prototype

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# LOS + LMS Prototype — Low-Level Design (LLD)

**Date:** 2026-08-05 **Companion docs:** [Spec](#) · [HLD](#) · [Plan](#)

## 1. Module map

Every file in the repo, with its responsibility and the symbols it exports. Tasks in the plan add to this list one at a time.

### `packages/shared/src/index.ts`

- **Responsibility:** Single source of truth for cross-tier types and validation.
- **Exports:**
  - `LoanStatus` = 'pending' | 'approved' | 'rejected' | 'disbursed'
  - `LoanLifecycle` = 'active' | 'closed'
  - `Employment` = 'salaried' | 'self\_employed'
  - `Role` = 'applicant' | 'admin'
  - `SignupInput` (Zod), `LoginInput` (Zod)
  - `ApplicationInput` (Zod): { amount: number (positive, ₹1k-₹10L), termMonths: enum [6,12,24,36], annualRateBps: enum [1000,1200,1500,1800,2000,2400], purpose: string 1..200, employment: Employment }
  - `DecisionInput` (Zod): { decision: 'approve' | 'reject', reason?: string 1..500 }
  - DTOs (inferred TS types): `UserDTO` , `ApplicationDTO` , `LoanDTO` , `InstallmentDTO`

### `apps/api/src/`

| File                       | Responsibility                                                                                         | Key exports                                                             |
|----------------------------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| <code>server.ts</code>     | Process entry. Loads env, builds app, calls <code>migrate()</code> + <code>seed()</code> then listens. | (default export)<br><code>start()</code>                                |
| <code>app.ts</code>        | Fastify factory. Registers plugins, decorators, error handler, static handler.                         | <code>buildApp()</code>                                                 |
| <code>db/client.ts</code>  | Initializes better-sqlite3 + Drizzle. Reads <code>DATABASE_URL</code> .                                | <code>db</code> (Drizzle instance),<br><code>sqlite</code> (raw handle) |
| <code>db/schema.ts</code>  | Drizzle table definitions. See §2.                                                                     | All table objects +<br>inferred row types                               |
| <code>db/migrate.ts</code> | Applies Drizzle migrations. Idempotent.                                                                | <code>migrate()</code>                                                  |

| File                   | Responsibility                                                                                                                                                         | Key exports                                                                                                                      |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| db/seed.ts             | Idempotent upsert of two applicants + one mid-flight loan. See §6.                                                                                                     | seed()                                                                                                                           |
| domain/amortization.ts | Pure functions:<br>calculateEmi(principal, annualRateBps, termMonths) ,<br>generateSchedule(principal, annualRateBps, termMonths, startDate) . See §4.                 | calculateEmi ,<br>generateSchedule                                                                                               |
| domain/underwriting.ts | Pure function:<br>evaluate(monthlyIncomeCents, principal, annualRateBps, termMonths) returns {<br>recommendation: 'approve'   'reject',<br>reason?: string } . See §5. | evaluate                                                                                                                         |
| lib/auth.ts            | signToken(userId) ,<br>verifyToken(token) ,<br>hashPassword(pw) ,<br>verifyPassword(pw, hash) , cookie helpers.                                                        | same                                                                                                                             |
| lib/errors.ts          | Typed domain error classes + Fastify error mapper. See §7.                                                                                                             | AppError ,<br>ValidationError ,<br>UnauthorizedError ,<br>ForbiddenError ,<br>NotFoundError ,<br>ConflictError ,<br>errorHandler |
| lib/money.ts           | Helpers for integer-cents math.<br>roundCents(n) , formatINR(cents) .                                                                                                  | same                                                                                                                             |
| routes/auth.ts         | POST /signup , POST /login , POST /logout , GET /me .                                                                                                                  | Fastify plugin                                                                                                                   |
| routes/applications.ts | POST /applications (creates pending) , GET /applications , GET /applications/:id .                                                                                     | Fastify plugin                                                                                                                   |

| File            | Responsibility                                                                                            | Key exports    |
|-----------------|-----------------------------------------------------------------------------------------------------------|----------------|
| routes/loans.ts | GET /loans , GET /loans/:id , POST /loans/:id/installments/:n/pay .                                       | Fastify plugin |
| routes/admin.ts | GET /admin/applications , POST /admin/applications/:id/decision , POST /admin/applications/:id/disburse . | Fastify plugin |

#### apps/web/src/

| File                     | Responsibility                                                                            | Key exports                                               |
|--------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| main.tsx                 | Mounts React app, wraps in QueryClientProvider + BrowserRouter .                          | (entry)                                                   |
| App.tsx                  | Top-level routes + <Toaster /> .                                                          | default                                                   |
| lib/api.ts               | apiFetch(path, opts) — same-origin fetch with credentials: 'include' , throws on non-2xx. | apiFetch                                                  |
| lib/format.ts            | formatINR(cents) , formatDate(iso) .                                                      | same                                                      |
| hooks/useMe.ts           | TanStack Query: ['me'] .                                                                  | useMe                                                     |
| hooks/useApplications.ts | TanStack Query: ['applications'] .                                                        | useApplications , useCreateApplication                    |
| hooks/useLoans.ts        | TanStack Query: ['loans'] and ['loans', id] .                                             | useLoans , useLoan , usePayInstallment                    |
| hooks/useAdmin.ts        | TanStack Query: ['admin', 'applications', status] .                                       | useAdminApplications , useDecideApplication , useDisburse |
| components/AuthGuard.tsx | Redirects to /login if no me ; to /dashboard if me.role !== 'admin' on /admin/* .         | default                                                   |
| components/Money.tsx     | Renders formatINR(value) .                                                                | default                                                   |

| File                                | Responsibility                                                                                           | Key exports |
|-------------------------------------|----------------------------------------------------------------------------------------------------------|-------------|
| components/StatusBadge.tsx          | Maps <code>LoanStatus</code> / <code>LoanLifecycle</code> to shadcn <code>&lt;Badge&gt;</code> variants. | default     |
| routes/landing.tsx                  | Hero + login/signup CTAs.                                                                                | default     |
| routes/login.tsx                    | RHF + Zod form. On success, navigate <code>/dashboard</code> .                                           | default     |
| routes/signup.tsx                   | RHF + Zod form, requires <code>monthlyIncome</code> .                                                    | default     |
| routes/apply.tsx                    | 3-step form (shadcn <code>Form</code> + simple stepper), submit creates <code>ApplicationDTO</code> .    | default     |
| routes/dashboard.tsx                | Lists <code>applications</code> and active <code>loans</code> , "Next EMI" card per loan.                | default     |
| routes/loan-detail.tsx              | Renders installments table, "Pay EMI" button on next unpaid row.                                         | default     |
| routes/admin/queue.tsx              | Lists <code>pending</code> applications, links to detail.                                                | default     |
| routes/admin/application-detail.tsx | Renders application + rule recommendation + Approve/Reject form + (if approved) Disburse button.         | default     |

## 2. Database schema (Drizzle, SQLite)

All money: `integer` (cents). All timestamps/dates: `text` (ISO 8601). All IDs: `text` (nanoid, 21 chars).

```
// apps/api/src/db/schema.ts
import { sqliteTable, text, integer, uniqueIndex } from 'drizzle-orm/sqlite-core';

export const users = sqliteTable('users', {
  id: text('id').primaryKey(),
  email: text('email').notNull().unique(),
  passwordHash: text('password_hash').notNull(),
  fullName: text('full_name').notNull(),
});
```

```

    role:          text('role', { enum: ['applicant', 'admin'] }).notNull(),
    monthlyIncome: integer('monthly_income').notNull(),    // cents
    createdAt:     text('created_at').notNull(),
  });

export const loanApplications = sqliteTable('loan_applications', {
  id:             text('id').primaryKey(),
  userId:         text('user_id').notNull().references(() => users.id),
  amountCents:    integer('amount_cents').notNull(),
  termMonths:     integer('term_months').notNull(),
  annualRateBps:  integer('annual_rate_bps').notNull(),
  purpose:        text('purpose').notNull(),
  employment:     text('employment', { enum: ['salaried', 'self-employed'] }).notNull(),
  status:         text('status', { enum: ['pending', 'approved', 'rejected', 'disbursed'] }),
  decisionReason: text('decision_reason'),
  decidedBy:      text('decided_by').references(() => users.id),
  decidedAt:      text('decided_at'),
  disbursedAt:    text('disbursed_at'),
  createdAt:      text('created_at').notNull(),
});

export const loans = sqliteTable('loans', {
  id:             text('id').primaryKey(),                                // = loanApp
  applicationId:  text('application_id').notNull().unique().references(() => loanApplica
  userId:         text('user_id').notNull().references(() => users.id),
  principalCents: integer('principal_cents').notNull(),
  annualRateBps:  integer('annual_rate_bps').notNull(),
  termMonths:     integer('term_months').notNull(),
  startDate:      text('start_date').notNull(),
  endDate:        text('end_date').notNull(),
  status:         text('status', { enum: ['active', 'closed'] }).notNull(),
  outstandingCents: integer('outstanding_cents').notNull(),
});

export const installments = sqliteTable('installments', {
  id:             text('id').primaryKey(),
  loanId:         text('loan_id').notNull().references(() => loans.id),
  sequence:       integer('sequence').notNull(),
  dueDate:        text('due_date').notNull(),
  principalDue:   integer('principal_due').notNull(),
  interestDue:    integer('interest_due').notNull(),
  paidAmount:     integer('paid_amount').notNull().default(0),
  paidAt:         text('paid_at'),
}, (t) => ({
  uniqLoanSeq: uniqueIndex('uniq_loan_seq').on(t.loanId, t.sequence),
}));

```

Inferred row types (used everywhere downstream):

```
type User          = typeof users.$inferSelect;
type NewUser       = typeof users.$inferInsert;
type LoanApplication = typeof loanApplications.$inferSelect;
type Loan          = typeof loans.$inferSelect;
type Installment   = typeof installments.$inferSelect;
```

### 3. API contracts

All routes return JSON. Errors: `{ error: string, message?: string, issues?: ZodIssue[] }`. Auth via cookie `lms_session` (httpOnly, SameSite=Lax). 401 no/invalid session, 403 wrong role, 404 not found / not owner, 409 domain conflict, 400 validation.

#### POST /api/auth/signup

- Body: SignupInput = { email, password (>=8), fullName, monthlyIncome (>=0) }
- 201: { user: UserDTO } + sets cookie
- 409: { error: 'EmailAlreadyUsed' } if email exists

#### POST /api/auth/login

- Body: LoginInput = { email, password }
- 200: { user: UserDTO } + sets cookie
- 401: { error: 'InvalidCredentials' }

#### POST /api/auth/logout

- 204, clears cookie

#### GET /api/auth/me

- 200: { user: UserDTO }
- 401: { error: 'Unauthorized' }

#### POST /api/applications

- Auth: any session.
- Body: ApplicationInput .
- Behavior: compute `evaluate(monthlyIncome, amount, rate, term)` ; if reject and `recommendation === 'reject'` , auto-set `status='rejected'` , `decision_reason = rule reason` , `decided_by = null` , `decided_at = now` . Else `status='pending'` . Returns the application with `recommendation` field.
- 201: { application: ApplicationDTO & { recommendation: { rule: 'approve' | 'reject', reason?: string } } }

#### GET /api/applications

- Auth: any session.
- 200: { applications: ApplicationDTO[] } (current user's only)

#### GET /api/applications/:id

- Auth: owner or admin.
- 200: { application: ApplicationDTO & { recommendation } }
- 404: not found / not owner (and not admin)

#### GET /api/loans

- Auth: any session. Applicants see own; admins see all.
- 200: { loans: LoanDTO[] }

#### GET /api/loans/:id

- Auth: owner or admin.
- 200: { loan: LoanDTO, installments: InstallmentDTO[] }

#### POST /api/loans/:id/installments/:n/pay

- Auth: owner.
- Behavior: in a transaction, find installment with `sequence = :n` and `paid_at IS NULL` ; if not found, rollback and 409. Else mark paid, decrement `loans.outstanding_cents` , if last installment set `loans.status='closed'` .
- 200: { installment: InstallmentDTO, loan: LoanDTO }
- 409: { error: 'NoInstallmentFound' } or { error: 'LoanClosed' }

#### GET /api/admin/applications?status=pending

- Auth: admin.
- 200: { applications: (ApplicationDTO & { applicant: UserDTO, recommendation })[] }

#### POST /api/admin/applications/:id/decision

- Auth: admin.
- Body: DecisionInput .
- Behavior: 404 if not found or not `pending` . 409 if not `pending` . UPDATE `status` , `decision_reason` , `decided_by = current user` , `decided_at = now` .
- 200: { application }

#### POST /api/admin/applications/:id/disburse



- Auth: admin.
- Behavior: in a transaction, 404 if not found, 409 if not `approved` . Generate `startDate = today` , `endDate = startDate + termMonths` . Compute `outstandingCents = principalCents` . INSERT `loans` (`id = application.id`). Run `generateSchedule` and bulk INSERT `installments` . UPDATE `loanApplications.status='disbursed'` , `disbursed_at=now` .
- 200: `{ loan, installments }`

## 4. Amortization ( `apps/api/src/domain/amortization.ts` )

```
export function calculateEmi(
  principalCents: number,
  annualRateBps: number,
  termMonths: number
): number; // integer cents, Math.round half-up

export interface ScheduleRow {
  sequence: number; // 1..N
  dueDate: string; // ISO yyyy-mm-dd, first = startDate + 1 month
  principalDue: number; // integer cents
  interestDue: number; // integer cents
}

export function generateSchedule(
  principalCents: number,
  annualRateBps: number,
  termMonths: number,
  startDate: string // ISO yyyy-mm-dd
): ScheduleRow[]; // length = termMonths, sum(principalDue) === principalCents
```

### Algorithm:

1. `r = annualRateBps / 10000 / 12` (monthly rate, decimal).
2. `emi = roundCents(principalCents * r * (1+r)^N / ((1+r)^N - 1))` . Special case `r === 0` : `emi = roundCents(principalCents / N)` .
3. For `i = 1..N-1` : `interestDue = roundCents(outstanding * r)` , `principalDue = emi - interestDue` , `outstanding -= principalDue` .
4. Final installment ( `i = N` ): `interestDue = roundCents(outstanding * r)` , `principalDue = outstanding` (forces the sum to match).
5. `dueDate = add i months to startDate` (using `date-fns/addMonths` with end-of-month clamp).

**Worked example** (₹50,000, 15% APR, 12 months, start 2026-01-01):

- `r = 0.0125` , `N = 12` , `emi ≈ 4,512.88` cents = ₹4,512.88.

- Final installment absorbs ~₹0.12 rounding.
- Schedule sums exactly to ₹50,000.00.

## 5. Underwriting ( `apps/api/src/domain/underwriting.ts` )

```
export interface UnderwritingResult {
  recommendation: 'approve' | 'reject';
  reason?: string;          // present iff reject
}

export function evaluate(
  monthlyIncomeCents: number,
  principalCents: number,
  annualRateBps: number,
  termMonths: number
): UnderwritingResult;
```

### Rules (in order, first match wins):

1. If `monthlyIncomeCents <= 0` → reject "Income not provided" .
2. Compute `emi = calculateEmi(...)` .
3. If `emi / monthlyIncomeCents > 0.50` → reject "EMI exceeds 50% of income" .
4. If `monthlyIncomeCents < 3 * emi` → reject "Income insufficient for requested EMI" .
5. Otherwise approve.

The application is **always** stored (either as `pending` or `rejected` ). When the rule recommends reject, the application is auto-rejected with `decided_by = null` so the admin queue only shows genuinely pending work. The admin can still see auto-rejected applications on the applicant side via `GET /api/applications` .

## 6. State machine

Implemented in `routes/admin.ts` via a single `transition(app, target)` helper. Allowed transitions:

| From   | To                    | Trigger                                      | Side effects                                                              |
|--------|-----------------------|----------------------------------------------|---------------------------------------------------------------------------|
| (none) | <code>pending</code>  | <code>POST /applications</code><br>(rule ok) | INSERT row                                                                |
| (none) | <code>rejected</code> | <code>POST /applications</code><br>(rule no) | INSERT row, set <code>decision_reason</code> ,<br><code>decided_at</code> |

| From     | To        | Trigger                   | Side effects                                                                                         |
|----------|-----------|---------------------------|------------------------------------------------------------------------------------------------------|
| pending  | approved  | admin decision<br>approve | set <code>decided_by</code> , <code>decided_at</code>                                                |
| pending  | rejected  | admin decision<br>reject  | set <code>decided_by</code> , <code>decided_at</code> ,<br><code>decision_reason</code>              |
| approved | disbursed | admin disburse            | INSERT <code>loans</code> , bulk INSERT<br><code>installments</code> , set <code>disbursed_at</code> |

Rejected applications are terminal. Disbursed applications are terminal at the application level; lifecycle continues on the `loans` row.

## 7. Error model

```
// apps/api/src/lib/errors.ts
export class AppError extends Error { abstract readonly status: number; readonly code: string }
export class ValidationError extends AppError { status = 400; code = 'ValidationError'; }
export class UnauthorizedError extends AppError { status = 401; code = 'Unauthorized'; }
export class ForbiddenError extends AppError { status = 403; code = 'Forbidden'; }
export class NotFoundError extends AppError { status = 404; code = 'NotFound'; }
export class ConflictError extends AppError { status = 409; code: string; } // subclass s

export function errorHandler(err, req, reply) {
  if (err instanceof ZodError) return reply.code(400).send({ error: 'ValidationError', issue: err.issues });
  if (err instanceof AppError) return reply.code(err.status).send({ error: err.code, message: err.message });
  req.log.error(err);
  return reply.code(500).send({ error: 'InternalError' });
}
```

Specific conflict codes: `EmailAlreadyUsed` , `InvalidStateTransition` , `NoInstallmentFound` , `LoanClosed` .

## 8. Test plan

Run with `node --test --import tsx apps/api/test/*.test.ts` . No test framework.

| File                              | Coverage                                                                                                                                                       |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>amortization.test.ts</code> | EMI matches closed-form for known inputs; schedule sums to principal; final absorbs rounding; <code>r=0</code> case; <code>addMonths</code> end-of-month clamp |

| File                              | Coverage                                                                                                                                                                                                                |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>underwriting.test.ts</code> | Each rejection rule triggers; boundary cases (FOIR exactly 0.50, income exactly 3× EMI); <code>income=0</code> rejected                                                                                                 |
| <code>auth.test.ts</code>         | Signup creates user + sets cookie; login round-trips; logout clears; <code>/me</code> 401 without cookie; duplicate email 409                                                                                           |
| <code>applications.test.ts</code> | Auto-rejected application returned to client; pending visible to admin; admin decision transitions state; disburse generates correct schedule and sets <code>outstanding_cents = principal</code> ; double-disburse 409 |
| <code>payment.test.ts</code>      | Pay next unpaid installment decrements outstanding; mark last → <code>loans.status='closed'</code> ; pay already-paid 409 ( <code>NoInstallmentFound</code> ); pay after close 409 ( <code>LoanClosed</code> )          |

Tests use `':memory:'` SQLite, no network, no seed file. Web: no tests.

## 9. Seed data

`db/seed.ts` upserts deterministically (idempotent on email):

| Email                      | Role      | monthlyIncome (cents) | Loan                                                                                    |
|----------------------------|-----------|-----------------------|-----------------------------------------------------------------------------------------|
| <code>admin@lms.dev</code> | admin     | 0                     | —                                                                                       |
| <code>alice@lms.dev</code> | applicant | 5,000,000 (₹50,000)   | none — fresh applicant                                                                  |
| <code>bob@lms.dev</code>   | applicant | 10,000,000 (₹100,000) | 5,000,000 cents / 12 mo / 15% APR, started 2026-05-01, 3 installments paid, 9 remaining |

Passwords: `admin123` , `alice123` , `bob123` (bcrypt-hashed on seed). `bob` 's paid installments: `paid_at = (startDate + i months)` , `paid_amount = principalDue + interestDue` .

## 10. Money & formatting

- **Boundary:** integer cents only, both directions.
- **Display:** `formatINR(cents) = new Intl.NumberFormat('en-IN', { style: 'currency', currency: 'INR' }).format(cents / 100)` .

- **Dates:** ISO `yyyy-mm-dd` for date-only fields, ISO 8601 with `z` for timestamps. Client: `Intl.DateTimeFormat` .
- 

**End of LLD.** Phased implementation tasks live in the [Plan](#).